

1. DISPLAY DATA

65 • `SELECT * FROM Customer;`
 66
 67 • `SELECT * FROM Account;`
 68
 69 • `SELECT * FROM Bank_Transaction;`
 70
 71
 72
 73

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | **Result Grid** | Form Editor | Field Types

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 22:10:50
2	10002	DEPOSIT	15000.00	2026-08-15 22:10:50
3	10003	WITHDRAW	20000.00	2026-08-15 22:10:50
4	10004	DEPOSIT	5000.00	2026-08-15 22:10:50
5	10005	WITHDRAW	10000.00	2026-08-15 22:10:50
NULL	NULL	NULL	NULL	NULL

Customer 2 Account 3 Bank_Transaction 4 x Apply Revert Cont

Output

Action Output

#	Time	Action	Message
✓ 14	22:10:50	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 15	22:11:11	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 16	22:11:11	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 17	22:11:11	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned

2. DEPOSIT MONEY

70
 71 • `START TRANSACTION;`
 72
 73 • `UPDATE Account`
 74 `SET Balance = Balance + 5000`
 75 `WHERE Account_No = 10001;`
 76
 77

Output

Action Output

#	Time	Action	Message
✓ 16	22:11:11	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 17	22:11:11	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 18	22:49:57	START TRANSACTION	0 row(s) affected
✓ 19	22:50:15	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

3. CHECK BALANCE

The screenshot shows the SQL Developer interface with the following SQL script:

```
72
73 • UPDATE Account
74   SET Balance = Balance + 5000
75   WHERE Account_No = 10001;
76
77 • SELECT *
78   FROM Account
79   WHERE Account_No = 10001;
80
```

The **Result Grid** displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

The **Output** pane shows the following actions:

#	Time	Action	Message
✓ 17	22:11:11	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 18	22:49:57	START TRANSACTION	0 row(s) affected
✓ 19	22:50:15	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 20	22:51:51	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

4. CHECK BALANCE AFTER COMMIT

The screenshot shows the SQL Developer interface with the following SQL script:

```
80
81 • COMMIT;
82
83 • SELECT *
84   FROM Account
85   WHERE Account_No = 10001;
```

The **Result Grid** displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

The **Output** pane shows the following actions:

#	Time	Action	Message
✓ 19	22:50:15	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 20	22:51:51	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 21	22:52:40	COMMIT	0 row(s) affected
✓ 22	22:52:53	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

ROLL BACK

4. WITHDRAW MONEY

```
86
87 • START TRANSACTION;
88
89 • UPDATE Account
90   SET Balance = Balance - 10000
91   WHERE Account_No = 10001;
92
93
94
95
96
97
98
99
```

Output

Action Output

#	Time	Action	Message
✓ 21	22:52:40	COMMIT	0 row(s) affected
✓ 22	22:52:53	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 23	22:54:29	START TRANSACTION	0 row(s) affected
✓ 24	22:54:48	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

6. CHECK BALANCE

```
74
93 • SELECT *
94   FROM Account
95   WHERE Account_No = 10001;
96
97
```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
* NULL	NULL	NULL	NULL	NULL

Account 7 x

Output

Action Output

#	Time	Action	Message
✓ 22	22:52:53	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 23	22:54:29	START TRANSACTION	0 row(s) affected
✓ 24	22:54:48	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 25	22:56:55	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

7. CHECK BALANCE AFTER ROLLBACK

SQL Editor:

```
99 • SELECT *
100 FROM Account
101 WHERE Account_No = 10001;
102
103
104
```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Output:

#	Time	Action	Message
✓ 24	22:54:48	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 25	22:56:55	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 26	22:57:54	ROLLBACK	0 row(s) affected
✓ 27	22:58:56	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

SAVE POINT

8. FIRST UPDATE

SQL Editor:

```
102
103 • START TRANSACTION;
104
105 • UPDATE Account
106 SET Balance = Balance + 5000
107 WHERE Account_No = 10001;
108
109
110
111
112
113
114
115
```

Output:

#	Time	Action	Message
✓ 27	22:58:56	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✗ 28	23:00:15	CREATE DATABASE madhulika_2520090056	Error Code: 1007. Can't create database 'madhulika_2520090056'; databas
✓ 29	23:00:24	START TRANSACTION	0 row(s) affected
✓ 30	23:00:54	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

9. SECOND UPDATE

```
108
109 • SAVEPOINT Deposit1;
110
111 • UPDATE Account
112   SET Balance = Balance - 3000
113   WHERE Account_No = 10002;
114
115
116
117
118
119
120
121
```

Conte

Output

Action Output

#	Time	Action	Message
✓ 29	23:00:24	START TRANSACTION	0 row(s) affected
✓ 30	23:00:54	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 31	23:02:15	SAVEPOINT Deposit1	0 row(s) affected
✓ 32	23:02:28	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

10. THIRD UPDATE

```
115 • SAVEPOINT Withdrawal1;
116
117 • UPDATE Account
118   SET Balance = Balance + 10000
119   WHERE Account_No = 10003;
120
121
122
123
124
125
126
127
128
```

Conte

Output

Action Output

#	Time	Action	Message
✓ 31	23:02:15	SAVEPOINT Deposit1	0 row(s) affected
✓ 32	23:02:28	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 33	23:03:33	SAVEPOINT Withdrawal1	0 row(s) affected
✓ 34	23:03:47	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

11. CHECK ACCOUNTS

121 • ROLLBACK TO SAVEPOINT Withdrawal1;
122
123 • COMMIT;
124
125 • SELECT *
126 FROM Account;
127

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	60000.00	Hyderabad
	10002	102	Savings	72000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Account 9 x Apply Revert Conte

Output

Action Output

#	Time	Action	Message
✓ 34	23:03:47	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 35	23:04:59	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
✓ 36	23:05:13	COMMIT	0 row(s) affected
✓ 37	23:05:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned

BANK TRANSFER

12. DEDUCT FROM SENDER

128 • START TRANSACTION;
129
130 • UPDATE Account
131 SET Balance = Balance - 10000
132 WHERE Account_No = 10001;
133
134
135
136
137
138
139
140
141

Output

Action Output

#	Time	Action	Message
✓ 36	23:05:13	COMMIT	0 row(s) affected
✓ 37	23:05:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 38	23:06:31	START TRANSACTION	0 row(s) affected
✓ 39	23:06:46	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

13. ADD TO RECEIVER

```
134 • UPDATE Account
135     SET Balance = Balance + 10000
136     WHERE Account_No = 10002;
137
138
139
140
141
142
143
144
145
146
147
```

Output

Action Output

#	Time	Action	Message
✓ 37	23:05:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 38	23:06:31	START TRANSACTION	0 row(s) affected
✓ 39	23:06:46	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 40	23:07:40	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

14. CHECK BOTH ACCOUNTS

```
138 • SELECT *
139     FROM Account
140     WHERE Account_No IN (10001,10002);
141
142
143
144
```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada
* NULL	NULL	NULL	NULL	NULL

Account 10 x

Apply Revert

Output

Action Output

#	Time	Action	Message
✓ 38	23:06:31	START TRANSACTION	0 row(s) affected
✓ 39	23:06:46	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 40	23:07:40	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 41	23:08:42	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned

15. FINAL BALANCES

The screenshot shows a database management tool interface. The top pane displays SQL code for a transaction. The middle pane shows the 'Result Grid' with two rows of account data. The bottom pane shows the 'Output' window with a log of executed SQL statements and their results.

```
142 • COMMIT;
143
144 • SELECT
145     Account_No,
146     Balance
147 FROM Account
148 WHERE Account_No IN (10001,10002);
149
150
```

Account_No	Balance
10001	50000.00
10002	82000.00

Output:

#	Time	Action	Message
✓ 40	23:07:40	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 41	23:08:42	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned
✓ 42	23:12:26	COMMIT	0 row(s) affected
✓ 43	23:12:38	SELECT Account_No, Balance FROM Account WHERE Account_No IN (10001,10002)	2 row(s) returned

TRANSFER USING ROLLBACK

16. DEDUCT MONEY

The screenshot shows a database management tool interface. The top pane displays SQL code for a transaction to deduct money. The bottom pane shows the 'Output' window with a log of executed SQL statements and their results.

```
150
151 • START TRANSACTION;
152
153 • UPDATE Account
154     SET Balance = Balance - 20000
155     WHERE Account_No = 10001;
156
157
158
159
160
161
162
```

Output:

#	Time	Action	Message
✓ 42	23:12:26	COMMIT	0 row(s) affected
✓ 43	23:12:38	SELECT Account_No, Balance FROM Account WHERE Account_No IN (10001,10002)	2 row(s) returned
✓ 44	23:13:59	START TRANSACTION	0 row(s) affected
✓ 45	23:14:15	UPDATE Account SET Balance = Balance - 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

17. ADD MONEY

```
157 • UPDATE Account
158 SET Balance = Balance + 20000
159 WHERE Account_No = 10002;
160
161
162
163
164
165
166
167
```

Conte

Output

Action Output

#	Time	Action	Message
✓ 43	23:12:38	SELECT Account_No, Balance FROM Account WHERE Account_No IN (1000...	2 row(s) returned
✓ 44	23:13:59	START TRANSACTION	0 row(s) affected
✓ 45	23:14:15	UPDATE Account SET Balance = Balance - 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 46	23:15:57	UPDATE Account SET Balance = Balance + 20000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

18. CHECK BALANCE

```
160
161 • ROLLBACK;
162
163 • SELECT *
164 FROM Account
165 WHERE Account_No IN (10001,10002);
166
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	82000.00	Vijayawada
*	NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Account 12 x

Apply

Revert

Conte

Output

Action Output

#	Time	Action	Message
✓ 45	23:14:15	UPDATE Account SET Balance = Balance - 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 46	23:15:57	UPDATE Account SET Balance = Balance + 20000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 47	23:16:42	ROLLBACK	0 row(s) affected
✓ 48	23:16:57	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned

ACID

19. ATOMICITY

```
167 • START TRANSACTION;
168
169 • UPDATE Account
170   SET Balance = Balance - 10000
171   WHERE Account_No = 10001;
172
173 • UPDATE Account
174   SET Balance = Balance + 10000
175   WHERE Account_No = 10002;
176
177 • COMMIT;
```

Output

Action Output

#	Time	Action	Message
✓ 49	23:25:33	START TRANSACTION	0 row(s) affected
✓ 50	23:25:33	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 51	23:25:33	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 52	23:25:33	COMMIT	0 row(s) affected

20. DURABILITY

```
179
180 • START TRANSACTION;
181
182 • UPDATE Account
183   SET Balance = Balance + 5000
184   WHERE Account_No = 10001;
185
186 • COMMIT;
```

Output

Action Output

#	Time	Action	Message
✓ 53	23:26:17	ROLLBACK	0 row(s) affected
✓ 54	23:30:03	START TRANSACTION	0 row(s) affected
✓ 55	23:30:03	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 56	23:30:03	COMMIT	0 row(s) affected

TRANSACTION ISOLATION LEVELS

21. CHECK CURRENT ISOLATION LEVEL

187
188 • `SELECT @@GLOBAL.transaction_isolation;`
189
190

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	@@GLOBAL.transaction_isolation
▶	REPEATABLE-READ

Result 13 x | Read Only | Conte

Output :
Action Output

#	Time	Action	Message
✓ 54	23:30:03	START TRANSACTION	0 row(s) affected
✓ 55	23:30:03	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 56	23:30:03	COMMIT	0 row(s) affected
✓ 57	23:31:30	SELECT @@GLOBAL.transaction_isolation LIMIT 0, 1000	1 row(s) returned

22. READ UNCOMMITTED

190 • `SET SESSION TRANSACTION ISOLATION LEVEL`
191 `READ UNCOMMITTED;`
192 • `START TRANSACTION;`
193 • `SELECT *`
194 `FROM Account`
195 `WHERE Account_No = 10001;`
196 • `COMMIT;`
197
198

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

Account_No	Customer_ID	Account_Type	Balance	Branch
▶ 10001	101	Savings	45000.00	Hyderabad
• HOLD	HOLD	HOLD	HOLD	HOLD

Account 14 x | Apply | Revert | Conte

Output :
Action Output

#	Time	Action	Message
✓ 58	23:32:47	SET SESSION TRANSACTION ISOLATION LEVEL READ UNCOMMITTED	0 row(s) affected
✓ 59	23:32:47	START TRANSACTION	0 row(s) affected
✓ 60	23:32:47	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 61	23:32:47	COMMIT	0 row(s) affected

23. READ COMMITTED

198 • SET SESSION TRANSACTION ISOLATION LEVEL
199 READ COMMITTED;
200 • START TRANSACTION;
201 • SELECT *
202 FROM Account
203 WHERE Account_No = 10001;
204 • COMMIT;
205
206

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 15 x Apply Revert

Output

Action Output

#	Time	Action	Message
62	23:34:00	SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED	0 row(s) affected
63	23:34:00	START TRANSACTION	0 row(s) affected
64	23:34:00	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
65	23:34:00	COMMIT	0 row(s) affected

24. REPEATABLE READ

205
206 • SET SESSION TRANSACTION ISOLATION LEVEL
207 REPEATABLE READ;
208 • START TRANSACTION;
209 • SELECT *
210 FROM Account
211 WHERE Account_No = 10001;
212 • COMMIT;
213

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 16 x Apply Revert

Output

Action Output

#	Time	Action	Message
66	23:35:01	SET SESSION TRANSACTION ISOLATION LEVEL REPEATABLE READ	0 row(s) affected
67	23:35:01	START TRANSACTION	0 row(s) affected
68	23:35:01	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
69	23:35:01	COMMIT	0 row(s) affected

25. SERIALIZABLE

```
214 • SET SESSION TRANSACTION ISOLATION LEVEL
215     SERIALIZABLE;
216 • START TRANSACTION;
217 • SELECT *
218     FROM Account
219     WHERE Account_No = 10001;
220 • COMMIT;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad

Account 17 x [Apply] [Revert] [Conte]

Output

Action Output

#	Time	Action	Message
70	23:36:03	SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE	0 row(s) affected
71	23:36:03	START TRANSACTION	0 row(s) affected
72	23:36:03	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
73	23:36:03	COMMIT	0 row(s) affected

26. DIRTY READ EXPERIMENT

```
222 • SET SESSION TRANSACTION ISOLATION LEVEL
223     READ UNCOMMITTED;
224 • START TRANSACTION;
225 • UPDATE Account
226     SET Balance = Balance + 20000
227     WHERE Account_No = 10001;
228
229
```

Conte

Output

Action Output

#	Time	Action	Message
73	23:36:03	COMMIT	0 row(s) affected
74	23:36:59	SET SESSION TRANSACTION ISOLATION LEVEL READ UNCOMMITTED	0 row(s) affected
75	23:36:59	START TRANSACTION	0 row(s) affected
76	23:36:59	UPDATE Account SET Balance = Balance + 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

27. SESSION -2

The screenshot shows a SQL Server Enterprise Manager interface. The top pane contains a query with the following text:

```
229 • SET SESSION TRANSACTION ISOLATION LEVEL
230 READ UNCOMMITTED;
231 • START TRANSACTION;
232 • SELECT Balance
233 FROM Account
234 WHERE Account_No = 10001;
235
```

The bottom pane shows the 'Results' grid with a single row of data:

Balance
65000.00

The right-hand side of the interface includes a 'Tools' pane with buttons for 'Result Grid', 'Form Editor', and 'Field Types'. Below the query window, the 'Output' pane is visible, showing a table of execution messages:

#	Time	Action	Message
76	23:36:59	UPDATE Account SET Balance = Balance + 20000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
77	23:38:26	SET SESSION TRANSACTION ISOLATION LEVEL READ UNCOMMITTED	0 row(s) affected
78	23:38:26	START TRANSACTION	0 row(s) affected
79	23:38:26	SELECT Balance FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

READ COMMITTED EXPERIMENT

28. SESSION-1

The screenshot shows a SQL Server Enterprise Manager interface. The top pane contains a query with the following text:

```
244 • SET SESSION TRANSACTION ISOLATION LEVEL
245 READ COMMITTED;
246 • START TRANSACTION;
247 • UPDATE Account
248 SET Balance = Balance + 10000
249 WHERE Account_No = 10001;
250
251
252
253
254
255
256
```

The bottom pane shows the 'Results' grid, which is currently empty. The right-hand side of the interface includes a 'Tools' pane with buttons for 'Result Grid', 'Form Editor', and 'Field Types'. Below the query window, the 'Output' pane is visible, showing a table of execution messages:

#	Time	Action	Message
82	23:39:25	COMMIT	0 row(s) affected
83	23:40:19	SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED	0 row(s) affected
84	23:40:19	START TRANSACTION	0 row(s) affected
85	23:40:19	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

29. SESSION-2

251 • SET SESSION TRANSACTION ISOLATION LEVEL
252 READ COMMITTED;
253 • START TRANSACTION;
254 • SELECT Balance
255 FROM Account
256 WHERE Account_No = 10001;
257

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Balance
75000.00

Account 20 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 85	23:40:19	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 86	23:41:15	SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED	0 row(s) affected
✓ 87	23:41:15	START TRANSACTION	0 row(s) affected
✓ 88	23:41:15	SELECT Balance FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

260 • SELECT Balance
261 FROM Account
262 WHERE Account_No = 10001;
263
264 • COMMIT;
265
266

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Balance
75000.00

Account 21 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 88	23:41:15	SELECT Balance FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 89	23:41:53	COMMIT	0 row(s) affected
✓ 90	23:42:06	SELECT Balance FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 91	23:42:06	COMMIT	0 row(s) affected

REPEATABLE READ EXPERIMENT

30. SESSION-1

266 • SET SESSION TRANSACTION ISOLATION LEVEL
267 • REPEATABLE READ;
268 • START TRANSACTION;
269 • SELECT Balance
270 FROM Account
271 WHERE Account_No = 10001;
272

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Balance
75000.00

Account 22 x Read Only Co

Output

Action Output

#	Time	Action	Message
91	23:42:06	COMMIT	0 row(s) affected
92	23:44:06	SET SESSION TRANSACTION ISOLATION LEVEL REPEATABLE READ	0 row(s) affected
93	23:44:06	START TRANSACTION	0 row(s) affected
94	23:44:06	SELECT Balance FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

31. SESSION-2

273 • START TRANSACTION;
274 • UPDATE Account
275 SET Balance = Balance + 5000
276 WHERE Account_No = 10001;
277 • COMMIT;
278
279
280
281
282

Output

Action Output

#	Time	Action	Message
94	23:44:06	SELECT Balance FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
95	23:44:56	START TRANSACTION	0 row(s) affected
96	23:44:56	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
97	23:44:56	COMMIT	0 row(s) affected

279 • `SELECT Balance`
 280 `FROM Account`
 281 `WHERE Account_No = 10001;`
 282 • `COMMIT;`
 283
 284
 285

Result Grid

Balance
80000.00

Account 23 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 96	23:44:56	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 97	23:44:56	COMMIT	0 row(s) affected
✓ 98	23:45:41	SELECT Balance FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 99	23:45:41	COMMIT	0 row(s) affected

SERIALIZABLE EXPERIMENT

32. SESSION-1

284 • `SET SESSION TRANSACTION ISOLATION LEVEL`
 285 `SERIALIZABLE;`
 286 • `START TRANSACTION;`
 287 • `SELECT *`
 288 `FROM Account`
 289 `WHERE Account_No = 10001;`
 290

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	80000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 24 x Apply Revert

Output

Action Output

#	Time	Action	Message
✓ 99	23:45:41	COMMIT	0 row(s) affected
✓ 100	23:46:41	SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE	0 row(s) affected
✓ 101	23:46:41	START TRANSACTION	0 row(s) affected
✓ 102	23:46:41	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

33. SESSION-2

```
291 • SET SESSION TRANSACTION ISOLATION LEVEL
292     SERIALIZABLE;
293 • START TRANSACTION;
294 • UPDATE Account
295     SET Balance = Balance + 5000
296     WHERE Account_No = 10001;
297
298
299
300
```

Output

Action Output

#	Time	Action	Message
✓ 102	23:46:41	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 103	23:47:49	SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE	0 row(s) affected
✓ 104	23:47:49	START TRANSACTION	0 row(s) affected
✓ 105	23:47:49	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

34. TRANSACTION COMMANDS

```
299
300 • START TRANSACTION;
301 • SAVEPOINT Savepoint_Name;
302 • ROLLBACK TO SAVEPOINT Savepoint_Name;
303 • RELEASE SAVEPOINT Savepoint_Name;
304 • COMMIT;
305
306
307
308
309
```

Output

Action Output

#	Time	Action	Message
✓ 114	23:51:04	SAVEPOINT Savepoint_Name	0 row(s) affected
✓ 115	23:51:04	ROLLBACK TO SAVEPOINT Savepoint_Name	0 row(s) affected
✓ 116	23:51:04	RELEASE SAVEPOINT Savepoint_Name	0 row(s) affected
✓ 117	23:51:05	COMMIT	0 row(s) affected

BANK TRANSACTION EXAMPLES

35. DEPOSIT TRANSACTION

```
306 • START TRANSACTION;
307 • UPDATE Account
308   SET Balance = Balance + 5000
309   WHERE Account_No = 10001;
310 • COMMIT;
```

311
312
313

Conte

Output

Action Output

#	Time	Action	Message
✓ 117	23:51:05	COMMIT	0 row(s) affected
✓ 118	23:51:40	START TRANSACTION	0 row(s) affected
✓ 119	23:51:40	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 120	23:51:40	COMMIT	0 row(s) affected

36. WITHDRAWAL TRANSACTION

```
312 • START TRANSACTION;
313 • UPDATE Account
314   SET Balance = Balance - 5000
315   WHERE Account_No = 10001;
316 • COMMIT;
```

317
318
319
320
321
322

Conte

Output

Action Output

#	Time	Action	Message
✓ 120	23:51:40	COMMIT	0 row(s) affected
✓ 121	23:54:02	START TRANSACTION	0 row(s) affected
✓ 122	23:54:02	UPDATE Account SET Balance = Balance - 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 123	23:54:02	COMMIT	0 row(s) affected

37. CANCEL WITHDRAWAL

```
318 • START TRANSACTION;
319 • UPDATE Account
320   SET Balance = Balance - 10000
321   WHERE Account_No = 10001;
322 • ROLLBACK;
323
324
325
326
327
328
329
330
```

Conte

Output

Action Output

#	Time	Action	Message
✓ 123	23:54:02	COMMIT	0 row(s) affected
✓ 124	23:55:09	START TRANSACTION	0 row(s) affected
✓ 125	23:55:09	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 126	23:55:09	ROLLBACK	0 row(s) affected

38. TRANSFER WITH SAVEPOINT

```
324 • START TRANSACTION;
325 • UPDATE Account
326   SET Balance = Balance - 10000
327   WHERE Account_No = 10001;
328 • SAVEPOINT AfterDebit;
329 • UPDATE Account
330   SET Balance = Balance + 10000
331   WHERE Account_No = 10002;
332 • COMMIT;
333
334
335
336
337
338
339
340
```

Conte

Output

Action Output

#	Time	Action	Message
✓ 128	00:00:48	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 129	00:00:48	SAVEPOINT AfterDebit	0 row(s) affected
✓ 130	00:00:48	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 131	00:00:48	COMMIT	0 row(s) affected